

10.2 DC: Potential Dividers

Question Paper

Course	CIEA Level Physics
Section	10. D.C. Circuits
Topic	10.2 DC: Potential Dividers
Difficulty	Medium

Time allowed: 40

Score: /27

Percentage: /100

Question 1a

A battery is connected in series with resistors P and Q, as shown in Fig 1.1.

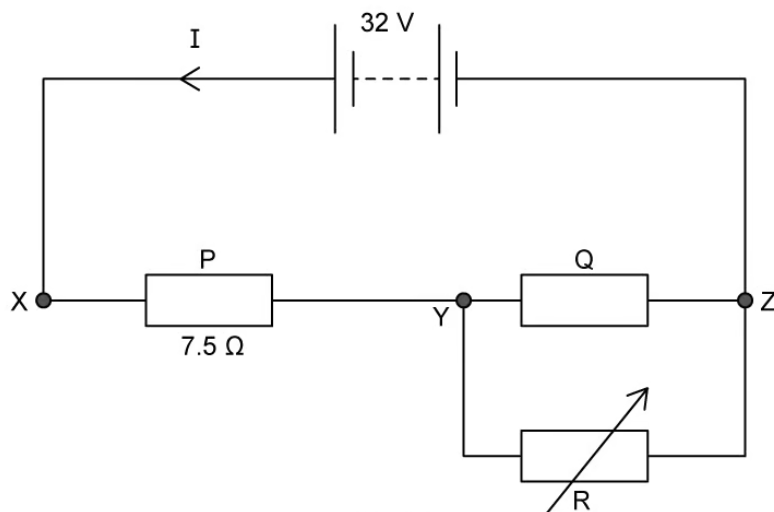


Fig.1.1

The resistance of P is 7.5Ω . The resistance of Q is constant. A variable resistor of resistance R is connected in parallel with Q.

The current I from the battery is changed by varying R from 5Ω to 20Ω . The variation with R of I is shown in Fig 1.2.

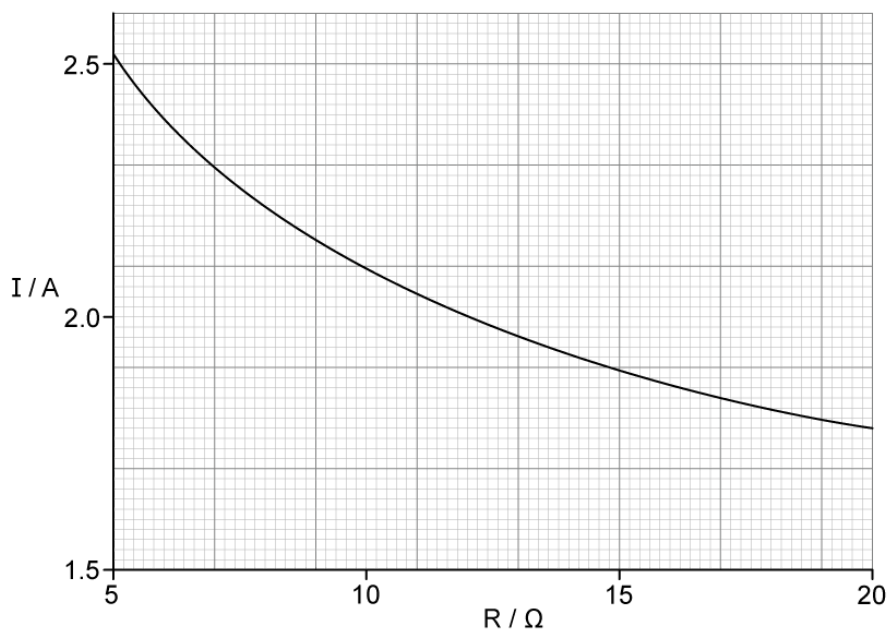


Fig.1.2

Use Fig 1.2 to state and explain the variation of the p.d. across resistor P as R is **decreased**. Numerical values are not required.

[2 marks]

Question 1b

For $R = 7.0\ \Omega$, calculate the p.d. between points Y and Z.

[2 marks]

Question 1c

Show that the resistance of Q is $95\ \Omega$.

[3 marks]

Question 1d

State and explain qualitatively how the power provided by the battery changes as the resistance R is decreased.

[1 mark]

Question 2a

Fig 1.1 shows a potential divider circuit with a battery of electromotive force (e.m.f.) 24.0 V connected to two resistors R_1 and R_2 .

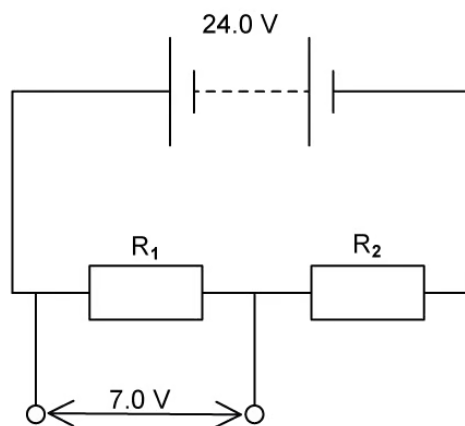


Fig 1.1

When the resistance of R_1 is $6.5\ \Omega$, the voltage across it is 7.0 V .

Show that the resistance of R_2 is $16\ \Omega$.

[2 marks]

Question 2b

The resistors are replaced with identical filament lamps A and B and the battery is connected in parallel to a linear potentiometer XY, as shown in Fig 1.2.

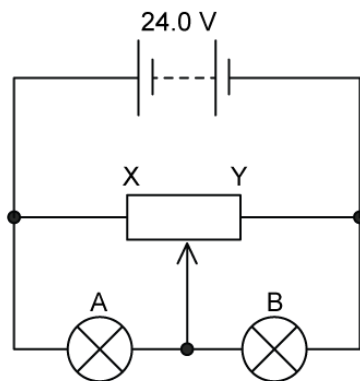


Fig 1.2

The lamps each have the same I - V characteristic shown in Fig. 1.3.

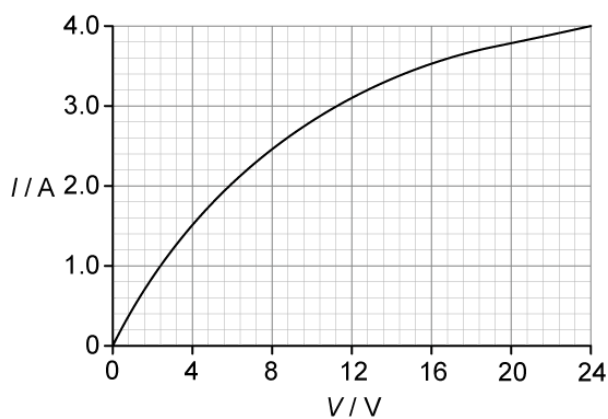


Fig 1.3

When the slider of the potentiometer is at its midpoint, as shown in Fig. 1.2, the current I in the battery is 4.12 A.

Determine the current in lamp B.

[1 mark]

Question 2c

Calculate the total resistance in the circuit.

[3 marks]

Question 2d

The slider of the potentiometer in (c) is moved to end X. State and explain the effect on the current of lamps A and B.

[2 marks]

Question 3a

A battery of negligible internal resistance is connected to a thermistor, a switch X and two fixed resistors, as shown in Fig. 1.1.

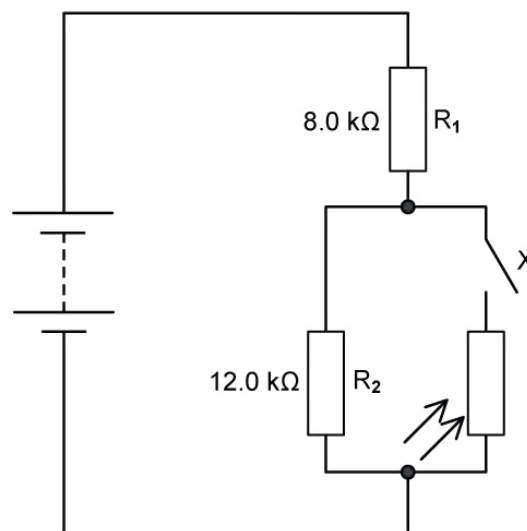


Fig 1.1

Resistor R_1 has resistance $8.0 \text{ k}\Omega$ and resistor R_2 has resistance $12.0 \text{ k}\Omega$. When switch X is open, the potential difference across R_2 is 18 V .

Show that the e.m.f of the battery is 30 V .

[2 marks]

Question 3b

Switch X is now closed. The current in the battery is 1.94 A.

Calculate the resistance of the LDR.

[3 marks]

Question 3c

The switch X in the circuit in (a) remains closed. The light intensity on the LDR increases.

By reference to the current in the battery, state and explain the effect, if any, of the increase in light intensity on the power produced by the battery.

[3 marks]

Question 3d

The light intensity on the LDR decreases.

State and explain what happens to the potential difference across R_1 .

[3 marks]



Model Answers: Medium

1a

(a) Use Fig 1.2 to state and explain the variation of the p.d. across resistor P as R is decreased

EITHER

- (As R decreases,) the current I increases; [1 mark]
- So the p.d. (across P) increases; [1 mark]

OR

- The parallel resistance of (of Q and R) decreases; [1 mark]
- The p.d. across the parallel resistors decreases, so p.d. across P increases; [1 mark]

[Total: 2 marks]

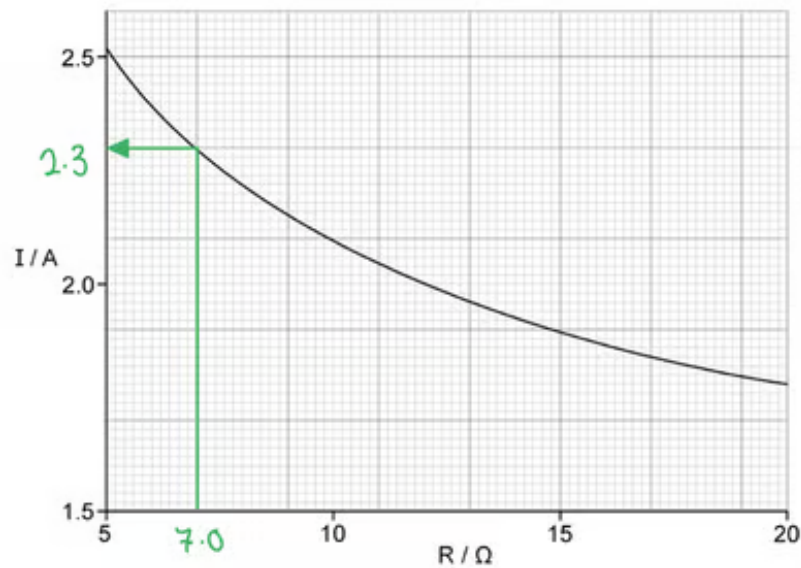
1b

(b) For $R = 7.0 \, \Omega$, calculate the p.d. between points Y and Z

List the known quantities:

- Resistance of variable resistor, $R = 7.0 \, \Omega$
- Resistance of resistor P, $R_P = 7.5 \, \Omega$
- E.m.f of the power supply, $V = 32 \, \text{V}$

Use the graph to determine the current I when $R = 7.0 \, \Omega$:



- $I = 2.3 \text{ A}$ [1 mark]

Method 1: Calculating p.d from current I

Calculate the p.d. across YZ:

- P.d across YZ = p.d across XZ – p.d. across XY
- P.d across YZ = $32 - (2.3 \times 7.5) = 14.75 = 15 \text{ V}$ [1 mark]

Method 2: Calculate p.d. from resistance

Determine the total resistance in the circuit, R_T :

- $R_T = \frac{V}{I} = \frac{32}{2.3} = 13.91 \Omega$

Calculate the p.d. across YZ:

- Resistance in Q-R parallel combination, $R_{QR} = 13.91 - 7.5 = 6.41 \Omega$
- P.d across YZ = $I \times R_{QR} = 2.3 \times 6.41 = 14.74 = 15 \text{ V}$ [1 mark]

[Total: 2 marks]

1c

(c) Show that the resistance of Q is 95Ω

List the known quantities:

- Resistance of variable resistor, $R = 7.0 \Omega$
- P.d. across YZ = 15 V (from part b)
- Current, $I = 2.3 \text{ A}$ (from part b)

Calculate the total resistance of Q and R parallel combination, R_T :

$$\bullet \quad R_T = \frac{V}{I} = \frac{15}{2.3} = 6.52 \Omega \text{ [1 mark]}$$

Calculate the resistance of resistor Q, R_Q :

$$\bullet \quad \frac{1}{R_T} = \frac{1}{R_Q} + \frac{1}{R}$$

$$\bullet \quad R_Q = \frac{1}{\frac{1}{R_T} - \frac{1}{R}} = \frac{1}{\frac{1}{6.52} - \frac{1}{7.0}} \text{ [1 mark]}$$

$$\bullet \quad R_Q = 95 \Omega \text{ [1 mark]}$$

[Total: 3 marks]

This question can also be done by calculating the current through Q, and then using

$$R = \frac{V}{I} \text{ to calculate the resistance through resistor Q.}$$

1d

(d) State and explain qualitatively how the power provided by the battery changes as the resistance R is decreased

- The power increases

AND

- Because the e.m.f is constant / power = $V \times$ current and the current (I) increases

$$\text{OR power} = \frac{V^2}{\text{resistance}} \text{ and the resistance decreases; [1 mark]}$$

[Total: 1 mark]

2a

(a) Show that the resistance of R_2 is $16\ \Omega$

List the known quantities:

- Resistance, $R_1 = 6.5\ \Omega$
- Voltage across R_1 , $V_{\text{out}} = 7.0\ \text{V}$
- E.m.f of the power supply, $V_{\text{in}} = 24.0\ \text{V}$

State the correct potential divider equation:

- $$V_{\text{out}} = V_{\text{in}} \times \left(\frac{R_1}{R_1 + R_2} \right)$$

Rearrange the potential divider equation for R_2 :

- $$R_2 = \left(\frac{R_1 \times V_{\text{in}}}{V_{\text{out}}} \right) - R_1$$

Substitute the values into the R_2 equation:

- $$R_2 = \left(\frac{6.5 \times 24.0}{7.0} \right) - 6.5 \text{ [1 mark]}$$
- $$R_2 = 15.786 = 16\ \Omega \text{ [1 mark]}$$

[Total: 2 marks]

You must be confident with rearranging the potential divider equation. If you are struggling with this one, have a look at the sketch below:

$$V_{out} = V_{in} \left(\frac{R_1}{R_1 + R_2} \right)$$

$$\frac{V_{out}}{V_{in}} = \left(\frac{R_1}{R_1 + R_2} \right) \quad \downarrow \div V_{in}$$

$$\frac{V_{out}}{V_{in}} (R_1 + R_2) = R_1 \quad \downarrow \times (R_1 + R_2)$$

$$R_1 + R_2 = \frac{R_1}{\frac{V_{out}}{V_{in}}} = \frac{R_1 V_{in}}{V_{out}} \quad \downarrow \div \frac{V_{out}}{V_{in}}$$

$$R_2 = \left(\frac{R_1 V_{in}}{V_{out}} \right) - R_1 \quad \downarrow -R_1$$

2b

(b) The current in lamp B is:

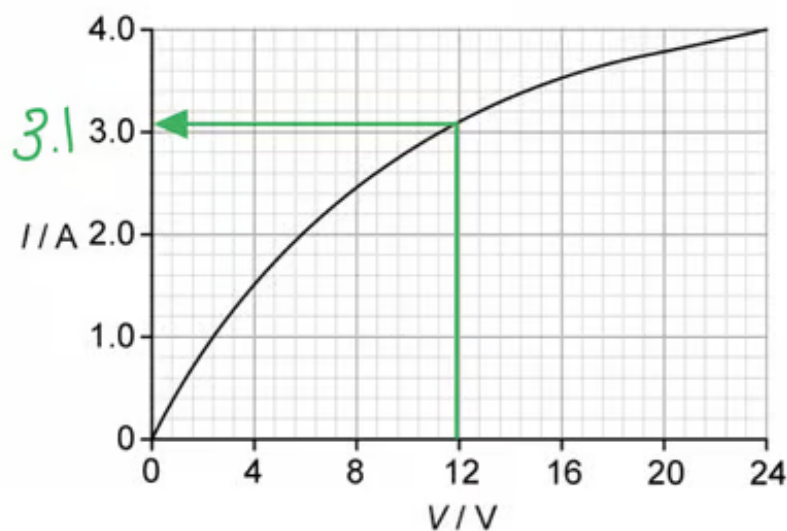
List the known quantities:

- Current in the battery, $I = 4.12 \text{ A}$
- E.m.f of the battery = 24.0 V

Determine the potential difference through lamp B:

- Since the slider is exactly in the middle, both lamps have a potential difference of $\frac{24.0}{2} = 12 \text{ V}$

Determine the current through lamp B at 12 V from the graph:



- 3.10 A [1 mark]

[Total: 1 mark]

2c

(c) Calculate the total resistance in the circuit

List the known quantities:

- Current through lamp B, $I_B = 3.10 \text{ A}$
- Current in the battery, $I = 4.12 \text{ A}$
- E.m.f of the battery, $V = 24.0 \text{ V}$

Calculate the resistance of the potentiometer between its ends X and Y, R_{XY} :

- Current through XY, $I_{XY} = I - I_B = 4.12 - 3.10 = 1.02 \text{ A}$

- $R_{XY} = \frac{V}{I_{XY}} = \frac{24.0}{1.02} = 23.529 \Omega \text{ [1 mark]}$

Calculate the resistance of each lamp, R_L :

- $R_L = \frac{V}{I} = \frac{12}{3.10} = 3.87 \Omega \text{ [1 mark]}$

Calculate the combined resistance, R_T :

- $R_{series} = R_L (\text{lamp A}) + R_L (\text{lamp B}) = 3.87 + 3.87 = 7.74 \Omega$

- $\frac{1}{R_T} = \frac{1}{R_{series}} + \frac{1}{R_{XY}} = \frac{1}{7.74} + \frac{1}{23.529}$

- $\frac{1}{R_T} = 0.1712$

- $R_T = 5.824 = 5.82 \Omega \text{ [1 mark]}$

[Total: 3 marks]

Read carefully which current the question is referring to. Since the potentiometer is connected in **parallel** with the lamps, the current from the **battery** will **split** between them (Kirchhoff's first law). Lamp A and B will have the same current through them.

2d

(d) State and explain the effect on the current of lamps A and B when the slider of the potentiometer is moved to end X

For lamp A:

- The p.d. across lamp A decreases to zero **AND** so the current is 0 [1 mark]

For lamp B:

- p.d. across lamp B increases to 24.0 V **AND** so the current is 4.0 A [1 mark]

[Total: 2 marks]

It is important you understand how the sliding contact works for a potentiometer. When it is at point X, **all** the p.d. from the power supply is available only to lamp B, so there is 0 for lamp A.

Where you **can** give values for a 'state and explain' question, make sure you do! In this case, just saying 'the p.d. across lamp B increases and so a current flows' will **not** be enough detail to obtain the marks, as you can state specific values.

The 4.0 A is read from the I-V graph when lamp B has a voltage of 24 V.

3a

(a) Show that the e.m.f of the battery is 30 V

List the known quantities:

- Potential difference across R_2 , $V_{out} = 18 \text{ V}$
- Resistance of R_1 , $R_1 = 8.0 \text{ k}\Omega$
- Resistance of R_2 , $R_2 = 12.0 \text{ k}\Omega$

State the potential divider equation:

$$V_{out} = V_{in} \times \left(\frac{R_2}{R_1 + R_2} \right)$$

Rearrange for the e.m.f of the battery, V_{in} and substitute in the values:

$$\bullet V_{in} = \frac{V_{out}}{\left(\frac{R_2}{R_1 + R_2}\right)} = \frac{18.0}{\left(\frac{12.0}{8.0 + 12.0}\right)} \quad [1 \text{ mark}]$$

$$\bullet V_{in} = 30 \text{ V} \quad [1 \text{ mark}]$$

[Total: 2 marks]

You do not need to convert from $\text{k}\Omega$ into Ω , since the fraction $\frac{R_2}{R_1 + R_2}$ will have no units (they cancel out) as long as they're all in $\text{k}\Omega$.

3b

(b) Calculate the resistance of the LDR.

List the known quantities:

- Resistance of R_1 , $R_1 = 8.0 \text{ k}\Omega$
- Resistance of R_2 , $R_2 = 12.0 \text{ k}\Omega$
- Current in the battery, $I = 1.94 \text{ mA} = 1.94 \times 10^{-3} \text{ A}$
- E.m.f of the battery, $V = 30 \text{ V}$ (from part a)

Calculate the total resistance in the circuit, R_T :

$$\bullet R_T = \frac{V}{I} = \frac{30}{1.94 \times 10^{-3}} = 15.46 \text{ k}\Omega \quad [1 \text{ mark}]$$

Calculate the resistance of the LDR, R_{LDR} :

$$\bullet 15.46 = R_{\text{parallel}} + 8.0 \Rightarrow R_{\text{parallel}} = 15.46 - 8.0 = 7.46 \text{ k}\Omega$$

$$\bullet \frac{1}{R_{\text{parallel}}} = \frac{1}{R_2} + \frac{1}{R_{\text{LDR}}} \Rightarrow \frac{1}{7.46} = \frac{1}{12.0} + \frac{1}{R_{\text{LDR}}}$$

$$\bullet R_{\text{LDR}} = \frac{1}{\frac{1}{7.46} - \frac{1}{12.0}} \quad [1 \text{ mark}]$$

$$\bullet R_{\text{LDR}} = 19.7 = 20 \text{ k}\Omega \quad [1 \text{ mark}]$$

[Total: 3 marks]

You should be confident in combining resistors in parallel and in series. It is better to set out the values algebraically before substituting in values.

3c

(c) State and explain the effect, if any, of the increase in light intensity on the power produced by the battery

- The LDR resistance decreases; [1 mark]
- (So the total resistance decreases) so the current increases (in the battery); [1 mark]
- Therefore, (since $P = IV$ and V is constant) so the power increases; [1 mark]

[Total: 3 marks]

3d

(d) If the light intensity on the LDR decreases:

- The resistance of the LDR increases; [1 mark]
- The combined resistance of the $12.0 \text{ k}\Omega / R_2$ and the LDR increases; [1 mark]

mark]

- Therefore, (since $P = IV$ and V is constant) so the power increases; [1 mark]

[Total: 3 marks]

3d

(d) If the light intensity on the LDR decreases:

- The resistance of the LDR increases; [1 mark]
- The combined resistance of the $12.0\text{ k}\Omega / R_2$ and the LDR increases; [1 mark]
- Therefore, the potential difference across R_1 decreases / R_1 receives a smaller share of the potential difference; [1 mark]

[Total: 3 marks]

The key to understanding potential divider questions is – if the resistance increases then so does the share of the potential difference (of the power supply) to that component.

